F175-SR-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM

F175-SW-701
SOUTHLAKES BOOSTER STATION
LV MOTOR CONTROL CENTRE

NORTHERN STAR
ALEGRASS LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

F175-GE-701
TEMPORARY GENERATOR
415V
200kVA
G 3~
400A
I >
1

415V FROM SMSB (BY OTHERS)
2

SWR EARTH BAR

F175GE701-DPS
GENERATOR DECONTACTOR PANEL

MCC EARTH BAR
NEUTRAL BAR
700/1A 2.5VA (PART OF ELR PACKAGE)
NOTE 1
I >

AFD
(NOTE 11)
400/ 1A 5VA Cl1
3
TEST
PM
2A 63A
20A 63A
2A 63A
PFR
2A 63A
MANUAL MCCB CHANGEOVER SWITCH
3 POLE
400A
400A
TEST
2A 63A
400/ 1A 2.5VA 5P20
3
400/ 1A 5VA Cl1.0
SD
NOTE 5
63A 100A
01F-07

415V 3NE~ 50Hz 400A 10kA for 1sec

01R-21	01R-05	01R-09	01R-13	01R-17
160A 250A	32A 160A	160A 250A	160A 250A	100A 160A
2A 20A	2A 20A	2A 20A	2A 20A	2A 20A
I > (NOTE 9)	I > (NOTE 9)	I > (NOTE 10)	I > (NOTE 10)	I > (NOTE 9)

ISL
UPS
AHF

10kW
F175-DB-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
AND AREA L&SP DB
2000-F175-SCM-EL-40013
2000-F175-TLD-EL-40001

10kW
F175-DB-702
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
240VAC UPS DB
2000-F175-TLD-EL-40002

F175UPS701-ISL
SOUTHLAKES BOOSTER STATION
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F175-SCM-EL-40014

F175-UPS-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
240VAC UPS
2000-F175-TLD-EL-40002

E-SHUT
F175SR701-LCS
NOTE 8

M 3~
75kW
F175-PW-701
SOUTHLAKES
BOOSTER PUMP 1
2000-F175-SCM-EL-40011

M 3~
75kW
F175-PW-702
SOUTHLAKES
BOOSTER PUMP 2
2000-F175-SCM-EL-40012

60A
F175-HF-701
SOUTHLAKES BOOSTER STATION
LV SWITCHROOM
ACTIVE HARMONIC FILTER
2000-F175-SCM-EL-40015

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC READING "M.E.N. LINK LOCATED IN F175-SW-701 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE". MEN LINK SHALL NOT BE INSTALLED AT THE GENERATOR.
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
4. ALL DRIVE I/O VIA COMMS NETWORK TO RTU. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F175-BLK-IC-40001.
5. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
6. MOULDED CASE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT UPSTREAM MCCB AND INCOMER REQUIREMENTS.
7. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
8. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/C WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.
9. COMPATIBLE 'TD' TOROID TO MATCH NHP ELR.
10. COMPATIBLE 'EL500S' TOROID TO MATCH AMPCONTROL ELV.
11. INCOMER ARC FAULT PROTECTION ZONE WILL TRIP UPSTREAM GENERATOR AND TRANSFORMER INCOMER ON LIGHT DETECTION ONLY. OUTGOING FEEDERS WILL TRIP TRANSFORMER INCOMER AND UPSTREAM GENSET ON CURRENT SUPERVISED LIGHT DETECTION.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 3, 4
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 3, 4
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 3, 4
- THERMAL OVERLOAD UNIT NOTE 7
- VARIABLE SPEED DRIVE NOTE 3, 4
- I > — EARTH LEAKAGE RELAY NOTE 3, 4
- FIELD-MOUNTED DECONTACTOR
- t° — MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- PFR — PHASE FAILURE RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN

HOLD

1. kVA RATING AND INTERNAL CB RATING OF GENERATOR.
2. NSR TO CONFIRM DETAILS OF NEW 415V SMSB BY OTHERS.

APPROVED FOR CONSTRUCTION

2000-F175-SLD-EL-40001_Rev0

Southlakes booster station F175-SW-701 LV motor control centre single line diagram